RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Friday, 14 August 2026, 2:17 PM
State	Finished
Completed on	Friday, 14 August 2026, 2:27 PM
Time taken	9 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, count = 0;
    scanf("%d", &n);

    count++;              // if condition: n == 1

    if (n == 1)
    {
        printf("*");
    }
    else
    {
        int i, j;

        count++;          // initialization of i
        for (i = 1; i <= n; i++)
        {
            count++;      // condition i <= n

            count++;      // initialization of j
            for (j = 1; j <= n; j++)
            {
                count++;  // condition j <= n
                count++;  // break
                break;
            }

            count++;      // increment of i
        }

                 // final condition i <= n
    }

    printf("%d", count);
```

```
38      return 0;
39  }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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